Algebra & Graphs

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A bag contains 6 green counters and 4 yellow counters. A counter is chosen at random, its colour recorded, and it is REPLACED. A second counter is then chosen. What is the probability that both counters are the same colour?

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A bank account pays compound interest of 3% per year. How many complete years will it take for an initial deposit of USD 2000 to exceed USD 3000?

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Port B is 40 km from Port A on a bearing of 120°. Port C is 50 km from Port A on a bearing of 030°. Calculate the distance between Port B and Port C.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A drawer contains 5 blue socks, 4 grey socks, and 3 black socks. Two socks are pulled at random without replacement. What is the probability of pulling two socks of different colours?

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In a circle of radius 10 cm, an arc subtends a central angle of 72°. Calculate the perimeter of this sector to 1 decimal place.

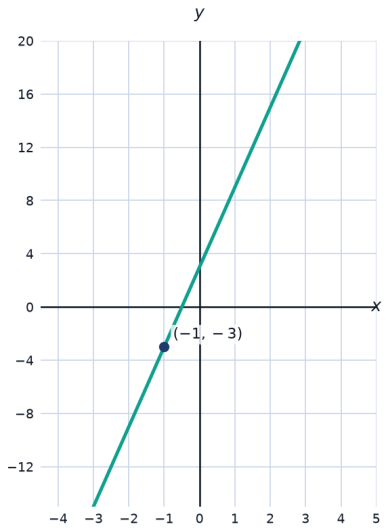
6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A vertical pole of height 12 m casts a shadow of length 9 m on horizontal ground. Calculate the angle of elevation of the sun to the nearest degree.

© ClipSAT — clipsat.org | All Rights Reserved

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

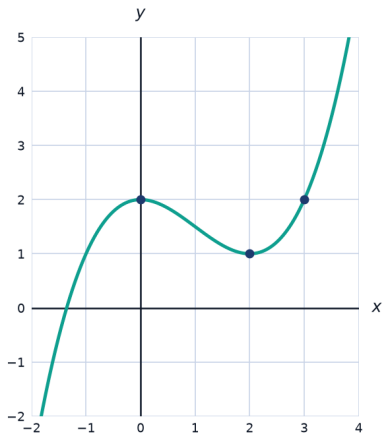
For the model $f(x) = 6x + 3$ in adding and subtracting fractions, evaluate $f(5)$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 2x + 4$ in fractions of a quantity, evaluate $f(2)$.



9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for fractions of a quantity consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the quadratic equation $2x^2 + 5x - 4 = 0$. Give your solutions correct to 2 decimal places.

© ClipSAT — clipsat.org | All Rights Reserved

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Algebra & Graphs · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The nth term of a sequence is $3n - 7$. (a) Find the 10th term. (b) Which term has value 23?

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the equation of the perpendicular bisector of the line segment joining $P(2, -1)$ and $Q(8, 7)$. Give your answer in the form $y = mx + c$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In triangle OAB, $\overrightarrow{OA} = 3\mathbf{a}$ and $\overrightarrow{OB} = 6\mathbf{b}$. Point P lies on AB such that $AP : PB = 1 : 2$. Find $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ in its simplest form.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Probability & Statistics

A solid toy is formed by attaching a cone of radius 4 cm and height 9 cm to the flat face of a solid hemisphere of radius 4 cm. Calculate the total volume of the toy. Give your answer correct to 3 significant figures.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 9.01 Functions

In triangle XYZ, $XY = 7$ cm, $YZ = 10$ cm, and angle $XYZ = 65^\circ$. Calculate (a) the length of XZ to 3 significant figures and (b) the area of triangle XYZ.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Number

A car travels a distance of $d = 180$ km correct to the nearest 10 km in a time of $t = 2.4$ hours correct to 1 decimal place. Calculate the upper bound for the average speed of the car in km/h.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Sets & Venn Diagrams

The table shows the time, t minutes, taken by 50 students to complete a homework task. Intervals: $0 < t \leq 10$ (freq: 8), $10 < t \leq 20$ (freq: 14), $20 < t \leq 30$ (freq: 18), $30 < t \leq 40$ (freq: 10). Calculate an estimate of the mean time taken.

© ClipSAT — clipsat.org | All Rights Reserved

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Coordinate Geometry

In a survey of 60 students, 35 study French, 28 study German, and 12 study both languages. Find (a) the number of students who study neither language, and (b) the probability that a student chosen at random studies French given that they study German.

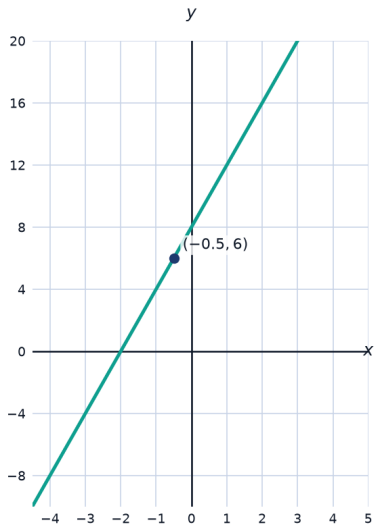
20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Mensuration

In a cyclic quadrilateral $ABCD$, angle $ABC = (2x + 15)^\circ$ and angle $ADC = (3x - 10)^\circ$. Find the value of x and hence the size of angle ABC. Give a geometric reason for your equation.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 8$ in name, identify, order, and compare fractions, evaluate $f(2)$.



a. 16

b. 20

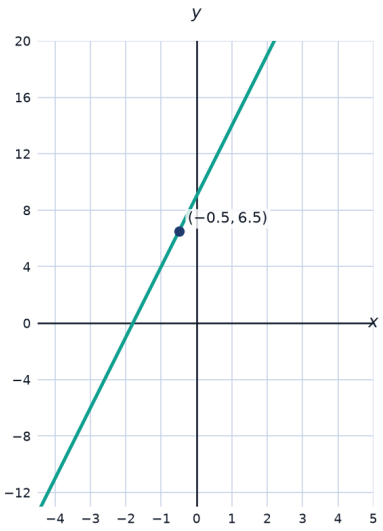
c. 8

d. 20

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 9$ in equivalent and simplified fractions, evaluate $f(2)$.



a. 19

b. 24

c. 10

d. 23

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for name, identify, order, and compare fractions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for equivalent and simplified fractions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
A data set recorded for adding and subtracting fractions consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5
b. 20.5
c. 21.5
d. 23.5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
A data set recorded for multiplying fractions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5
b. 22.5
c. 23.5
d. 25.5

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
For the model $f(x) = 3x + 5$ in multiplying fractions, evaluate $f(3)$.

a. 14
b. 17
c. 9
d. 18

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
A data set recorded for fractions and division consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5
b. 23.5
c. 24.5
d. 26.5

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

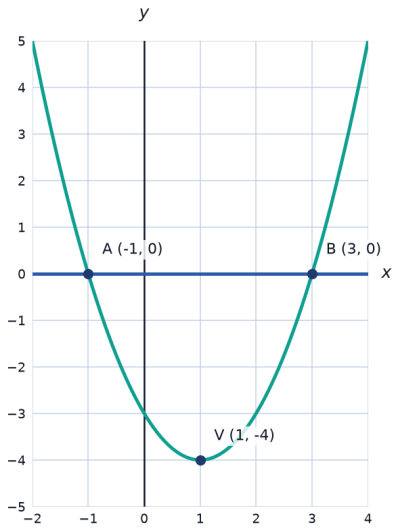
29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
For the model $f(x) = 4x + 6$ in fractions and division, evaluate $f(3)$.

a. 18
b. 22
c. 12
d. 22

© ClipSAT — clipsat.org | All Rights Reserved

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The diagram shows a curve with equation $y = ax^2 + bx + c$ and a straight line with equation $y = mx + d$. The curve and the line intersect at points A and B. The vertex of the parabola is at point V. The x-coordinates of the points A, B, and V are given as $x_A = -1$, $x_B = 3$, and $x_V = 1$ respectively. The y-coordinate of the vertex V is $y_V = -4$. Determine the value of a.



a. 1

b. 2

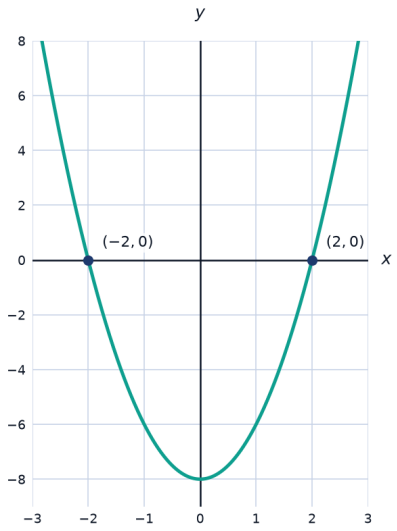
c. -1

d. -2

© ClipSAT — clipsat.org | All Rights Reserved

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The equation $y = 2x^2 - 8$ crosses the x-axis at:



a. $x = \pm 2$

b. $x = \pm 4$

c. $x = 2$

d. $x = \pm\sqrt{8}$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Solve the simultaneous equations $x + y = 7, x - y = 3$.

a. $x = 4, y = 3$

b. $x = 5, y = 2$

c. $x = 3, y = 4$

d. $x = 2, y = 5$

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve $2x^2 + 6x - 20 = 0$ using the quadratic formula. Give the larger root, as a fraction if needed.

a. -5

b. 2

c. 1

d. 3

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the nth term of the sequence $2, 6, 10, 14, \ldots$

a. $4n$

b. $4n - 4$

c. $5n - 2$

d. $4n - 2$

© ClipSAT — clipsat.org | All Rights Reserved

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve the simultaneous equations: $x + y = 9, x - y = -5$. Find x.

a. 9

b. 2

c. -6

d. 0

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A right triangular prism has a cross-section of a right triangle with legs 5 cm and 12 cm. The length of the prism is 20 cm. What is its volume?

a. 600 cm³

b. 1200 cm³

c. 300 cm³

d. 450 cm³

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A triangular prism has a cross-sectional (triangle) area of 24 cm^2 and length 4 cm. Find its volume.

a. 99 cm^3

b. 98 cm^3

c. 96 cm^3

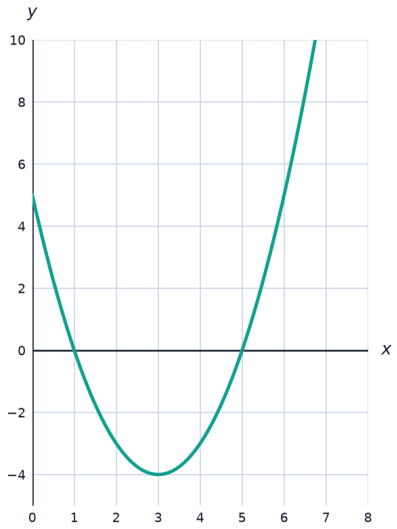
d. 97 cm^3

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Sets & Venn Diagrams

A function is defined by $f(x) = x^2 - 6x + 5$ for $x \geq 3$. \\ The graph of $y = f(x)$ is a parabola. A straight line with equation $y = k$ intersects the graph of $y = f(x)$ at exactly one point. \\ Find the possible value(s) of k.



a. $k = -4$

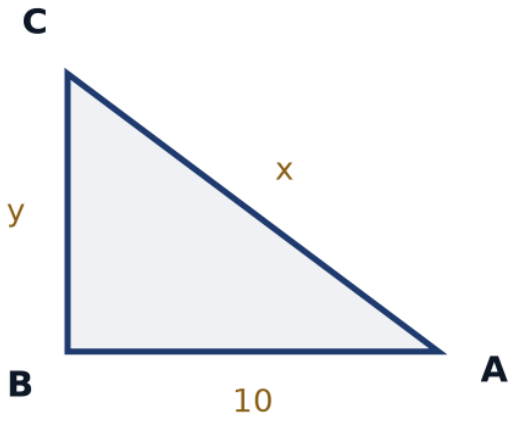
b. $k = 5$

c. $k = -4$ or $k = 5$

d. $k \geq 5$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Coordinate Geometry

In triangle ABC, angle ABC = 90°, AB = 10 cm, and angle BAC = θ. Given that $\sin\theta = \frac{3}{5}$, find the length of AC.



a. 16.7 cm

b. 8.0 cm

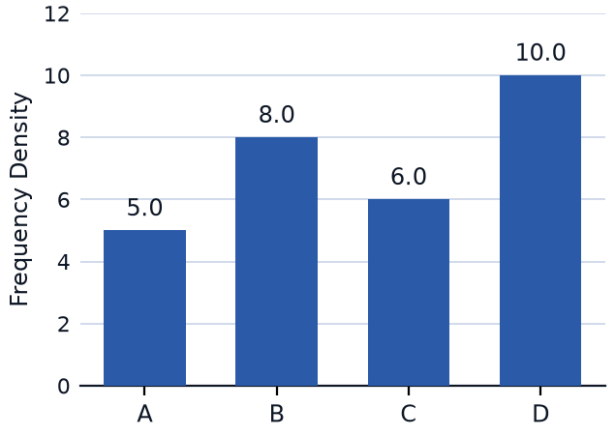
c. 12.5 cm

d. 6.0 cm

© ClipSAT — clipsat.org | All Rights Reserved

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Mensuration

On a histogram, a class interval is $20 \le x < 30$ with a frequency of 25. What is the frequency density for this bar?



a. 2.50

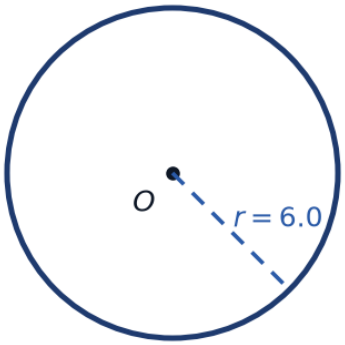
b. 5.00

c. 25

d. 10

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Geometry

A circle has radius 6 cm. Calculate the area of a sector with central angle 60° in terms of π.



a. 6π cm^2

b. 12π cm^2

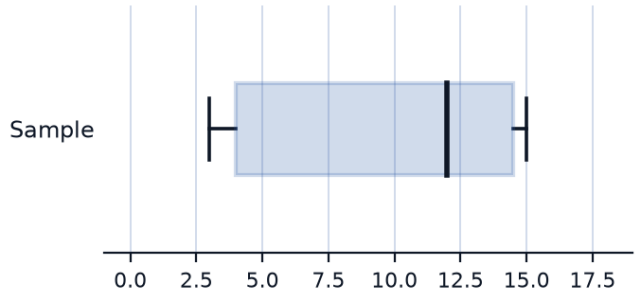
c. 3.0π cm^2

d. 360π cm^2

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Trigonometry

Find the range of the data set: $3, 5, 12, 14, 15$.



a. 18

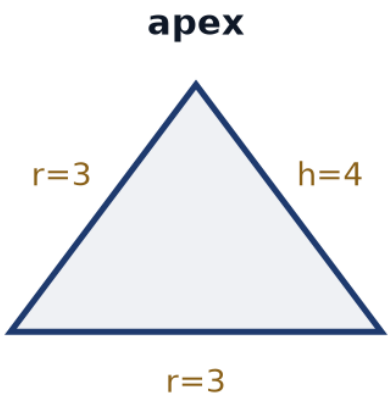
b. 7

c. 12

d. 6

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Transformations & Vectors

Volume of a cone with radius 3 and height 4:



a. 12π

b. 16π

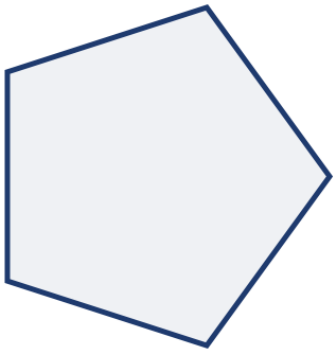
c. 36π

d. 48π

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Probability & Statistics

Interior angle of a regular pentagon:



a. 100°

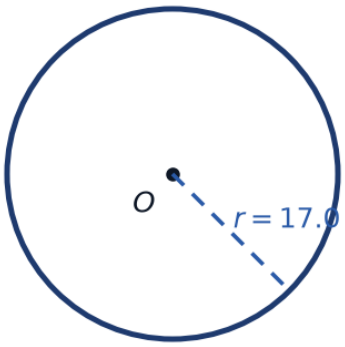
b. 108°

c. 112°

d. 120°

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Functions

A circle has radius 17 cm. A chord has length 30 cm. What is the perpendicular distance from the centre of the circle to the chord?



a. 8 cm

b. 7 cm

c. 9 cm

d. 10 cm

© ClipSAT — clipsat.org | All Rights Reserved